

Questions from Previous Years (Printer Topic)

Year / Exam	Question (Copy Exactly)	Matching Topic in printer.pdf / printer-2.pdf
Final Examination 2021	5 [D.] Make a comparison between buffers and spoolers.	Printer and Scanner Interfaces (Data Management)
Final Examination 2021	6 [D.] An interface incorporates several components including its communication type as well as the interface software. List out eight major communication types in case of a printer.	Printer and Scanner Interfaces
Final Examination 2020	1 [B.] What is Impact Printers? How does Impact Printers work? Explain with functional diagram.	2.1 Impact Printers
Final Examination 2016	5 (c) Write down the steps involved in starting a printing sequence using a printer controller chip.	Printer and Scanner Interfaces
Final Examination 2015	1 (b) Define interfaces. What are the characteristics of computer peripherals?	Printer and Scanner Interfaces
Final Examination 2015	1 (c) Differentiate between buffers and spoolers.	Printer and Scanner Interfaces (Data Management)
Final Examination 2014	1 (b) Define interfaces. What are the characteristics of computer peripherals?	Printer and Scanner Interfaces
Undated Final (from history)	How does inkjet printer work? Explain with figure.	2.4 Inkjet Printers

Undated Final (from history)	What do you mean by impact and non-impact printer?	2. Types of Printers / 2.1 Impact Printers
Undated Final (from history)	How does laser printer work? Explain with functional diagram.	2.5 Laser Printers
Undated Final (from history)	What is 82C55 PPI? What will be the operation of stepper motor when it is connected with 82C55 PPI?	Components inside a scanner (Stepper Motor) / Dot-Matrix printer problems

Total number of Printer-related questions found: 11

An interface incorporates several components including its communication type as well as the interface software. List out eight major communication types in case of a printer.

Solution:

Answer:

Eight major communication (interface) types used in printers are:

1. **Serial Interface (RS-232)**
Data is transmitted one bit at a time. Used in older printers.
2. **Parallel Interface (Centronics / IEEE 1284)**
Transfers multiple bits simultaneously. Common in older desktop printers.
3. **USB (Universal Serial Bus)**
The most common wired interface for modern printers. Supports fast data transfer.
4. **Ethernet (LAN)**
Connects the printer to a local network so multiple users can share it.
5. **Wi-Fi (Wireless LAN)**
Allows wireless printing from computers, smartphones, and tablets.
6. **Bluetooth**
Provides short-range wireless communication for printing from nearby devices.
7. **Infrared (IrDA)**
Uses infrared light for wireless communication. Found in some older portable printers.
8. **IEEE 1394 (FireWire)**
A high-speed communication interface used with some professional printers.

1 [B.] What is Impact Printers? How does Impact Printers work? Explain with functional diagram.

2. Impact Printers.

এগুলো অনেকটা টাইপরাইটারের মতো কাজ করে। প্রিন্টারের হেড বা চেইন কাগজের ওপর আঘাত (Impact) করার মাধ্যমে কালি বা কালির ফিতা থেকে লেখা ফুটিয়ে তোলে। এগুলো সাধারণত কিছুটা শব্দ বেশি করে।

Based on the image provided, here is the English translation of the highlighted Bengali text:
Impact Printers

"These work a lot like typewriters. The printer's head or chain strikes (Impacts) onto the paper to make writing appear from the ink or ink ribbon. These usually make a bit more noise."

How Impact Printers Work

Impact printers operate on a mechanical "strike-and-ink" principle:

1. **Data Reception:** The printer receives character or image data from the computer interface.
2. **Mechanical Striking:** A mechanism—such as a series of small metal pins in a printhead or characters positioned on a wheel—is driven forward mechanically.
3. **Ribbon Impression:** This striking mechanism forcefully hits an **inked ribbon** placed directly in front of the page.
4. **Transfer to Paper:** The impact transfers ink from the ribbon onto the paper, leaving behind a crisp imprint of a specific character or a pattern of dots.

ইমপ্যাক্ট প্রিন্টার যেভাবে কাজ করে (How Impact Printers Work)

ইমপ্যাক্ট প্রিন্টারগুলো একটি যান্ত্রিক "আঘাত এবং কালি" (strike-and-ink) নীতির ওপর ভিত্তি করে কাজ করে:

1. ডাটা গ্রহণ (**Data Reception**): প্রিন্টারটি কম্পিউটার ইন্টারফেস থেকে ক্যারেক্টার বা ইমেজের ডাটা গ্রহণ করে।
2. যান্ত্রিক আঘাত (**Mechanical Striking**): একটি যান্ত্রিক অংশ—যেমন প্রিন্টহেডের কিছু ছোট মেটাল পিন বা একটি হুইল বা চাকার গায়ে থাকা ক্যারেক্টার—যান্ত্রিকভাবে সামনের দিকে চালিত হয়।
3. রিবনে চাপ দেওয়া (**Ribbon Impression**): এই আঘাতকারী অংশটি কাগজের ঠিক সামনে থাকা একটি কালির রিবনে (inked ribbon) সজোরে আঘাত করে।
4. কাগজে স্থানান্তর (**Transfer to Paper**): এই আঘাতের ফলে রিবন থেকে কালি কাগজে স্থানান্তরিত হয়, যার ফলে নির্দিষ্ট ক্যারেক্টার বা ডটের (বিন্দুর) একটি স্পষ্ট ছাপ কাগজে বসে যায়।

Types of Impact Printers

1. Daisy-Wheel Printer

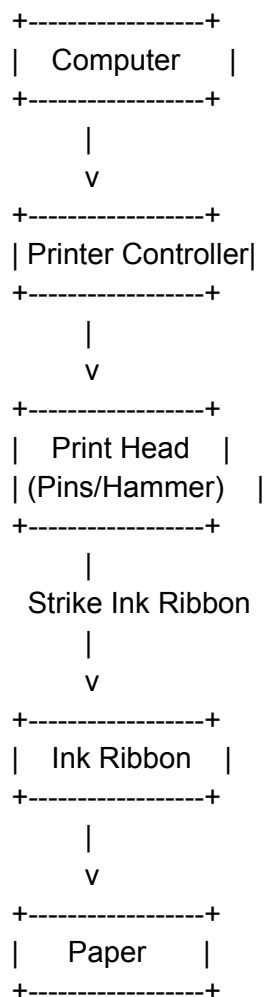
- Uses a **wheel-shaped disk** with embossed characters.
- A **hammer strikes the ink ribbon** to print characters on paper.

- Produces **letter-quality text** but **cannot print graphics**.
- **Slow and noisy**.

2. Dot-Matrix Printer

- Uses a **print head with multiple pins** (9 or 24 pins).
- The **pins strike an ink ribbon** to form characters and images using dots.
- Can print **text, graphics, and multi-part (carbon copy) forms**.
- **Noisy** but **low-cost and durable**.

Functional diagram :



Your uploaded PDF does **not** contain this specific question, so here's the standard exam answer.

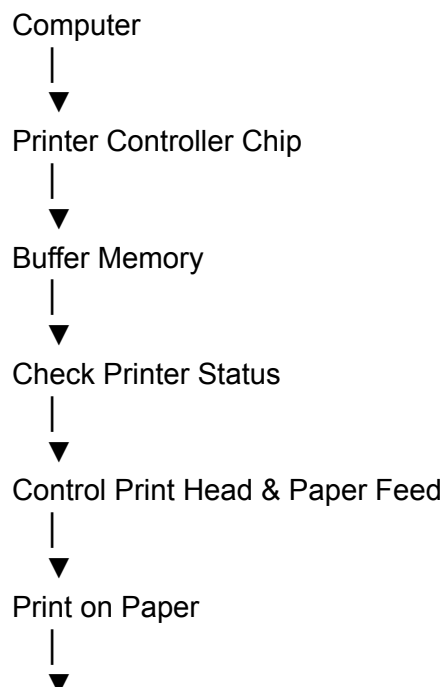
5(c) Write down the steps involved in starting a printing sequence using a printer controller chip. (5 Marks)

Answer

The steps involved in starting a printing sequence using a printer controller chip are:

1. **Receive Print Data**
The computer sends the print data and commands to the printer controller chip.
2. **Store Data in Buffer**
The controller chip temporarily stores the received data in the printer's memory (buffer).
3. **Check Printer Status**
The controller checks whether the printer is online, has paper, ink/toner, and is ready to print.
4. **Control the Print Mechanism**
The controller sends control signals to the print head, paper feed motor, and other printer components.
5. **Start Printing**
The print head prints the data on the paper while the paper feed mechanism moves the paper line by line.
6. **Complete the Job**
After all data has been printed, the controller sends a completion signal to the computer, indicating that the print job is finished.

Flow Diagram



Job Completed

Exam Tip

For a **5-mark question**, write:

- **6 steps** (4 marks)
- **Simple flow diagram** (1 mark)

This answer is concise, easy to memorize, and suitable for university exams.

Q. How does an Inkjet Printer work? Explain with figure. (6 Marks)

Answer

Definition

An **Inkjet Printer** is a **non-impact printer** that produces text and images by **spraying tiny droplets of ink through microscopic nozzles onto paper**. It provides high-quality color printing and operates quietly.

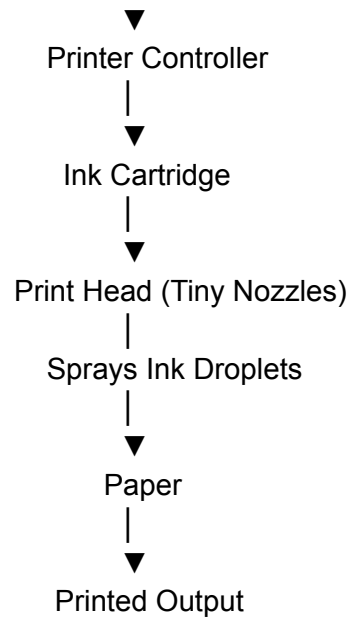
Working Principle

The working process of an Inkjet Printer is as follows:

1. The **computer sends print data** to the printer.
 2. The **printer controller** processes the data and sends instructions to the print head.
 3. Ink from the **ink cartridge** is supplied to the print head.
 4. The print head contains **tiny nozzles** that spray microscopic ink droplets onto the paper.
 5. The **paper feed mechanism** moves the paper line by line while the print head moves across it.
 6. Thousands of tiny ink droplets combine to form **text, graphics, and high-quality images**.
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Functional Diagram

Computer
|



Advantages

- High-quality text and photo printing.
 - Quiet operation.
 - Fast printing speed.
 - Suitable for home and office use.
-

Exam Tip (6 Marks)

- **Definition:** 1 Mark
- **Working Principle:** 4 Marks
- **Functional Diagram:** 1 Mark

This answer is concise, well-structured, and appropriate for a **6-mark university examination**.

Q. What do you mean by Impact and Non-Impact Printer? (4 Marks)

Impact Printer

An **Impact Printer** is a printer that produces text or images by **physically striking an inked ribbon against the paper**. These printers are generally **noisy** and are suitable for printing multi-part (carbon copy) forms.

Examples: Dot-Matrix Printer, Daisy-Wheel Printer.

Non-Impact Printer

A **Non-Impact Printer** is a printer that produces text or images **without physically striking the paper**. It uses technologies such as **ink spraying, laser beams, or heat**, making it **quiet, faster, and capable of producing high-quality output**.

Examples: Inkjet Printer, Laser Printer, Thermal Printer.

Difference Between Impact and Non-Impact Printer

Impact Printer	Non-Impact Printer
Prints by striking an ink ribbon against paper.	Prints without striking the paper.
Noisy operation.	Quiet operation.
Lower print quality.	High print quality.
Can print carbon copies.	Cannot print carbon copies.
Example: Dot-Matrix, Daisy-Wheel.	Example: Inkjet, Laser, Thermal.

Impact Printer

ইঙ্ক রিবনে আঘাত করে প্রিন্ট করে।

কাজ করার সময় বেশি শব্দ হয়।

প্রিন্টের মান তুলনামূলক কম।

Non-Impact Printer

কাগজে আঘাত না করে প্রিন্ট করে।

খুব কম শব্দ হয়।

প্রিন্টের মান অনেক ভালো।

কার্বন কপি (Multi-part form) প্রিন্ট করতে পারে।

কার্বন কপি প্রিন্ট করতে পারে না।

উদাহরণ: Dot-Matrix, Daisy-Wheel।

উদাহরণ: Inkjet, Laser, Thermal।

Q. How does a Laser Printer work? Explain with functional diagram. (6 Marks)

Answer

Definition (1 Mark)

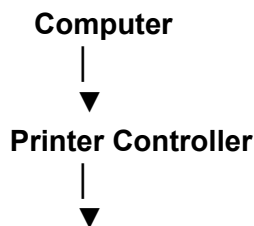
A Laser Printer is a non-impact printer that uses a laser beam and static electricity to create images on a photosensitive drum. The toner is then transferred and permanently fused onto paper using heat and pressure. It produces high-speed, high-quality prints.

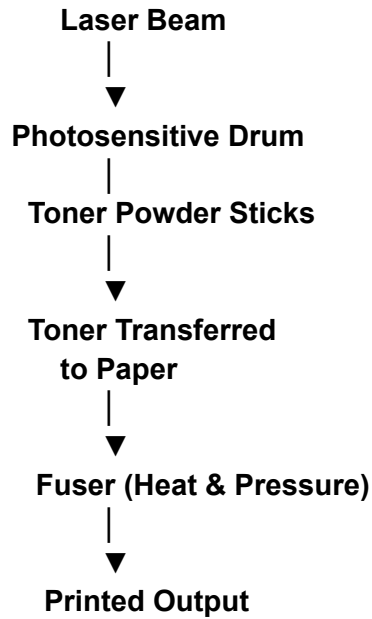
Working Principle (4 Marks)

The working process of a laser printer is as follows:

1. The computer sends print data to the printer.
 2. A laser beam scans the photosensitive drum, creating an electrostatic image on its surface.
 3. Toner powder sticks only to the charged areas of the drum.
 4. A sheet of paper passes over the drum, and the toner image is transferred onto the paper.
 5. The paper then passes through the fuser unit, where heat and pressure permanently fuse the toner onto the paper.
 6. Finally, the printed paper comes out, and the drum is cleaned for the next print job.
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Functional Diagram (1 Mark)





Advantages

- High-speed printing.
- Excellent print quality.
- Quiet operation.
- Suitable for large-volume printing.

Exam Tip (6 Marks)

- Definition: 1 Mark
- Working Principle: 4 Marks
- Functional Diagram: 1 Mark

This answer is suitable for writing in a 6-mark university examination.